

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

Chapter 4

DEMAND

Demand is a very crucial concept in economics. The determination of prices of various goods, services and resources in a free market economy depend upon two opposite forces "demand" and "supply". In order to highlight the problem of price determination we initially take into account the concept of demand as under:-

DEFINITION OF DEMAND:

Commonly demand means a mere "wish" or "desire" to have a commodity. But in economics at least two conditions must be fulfilled for the existence of demand.

- (a) A strong "will" to purchase any thing.
- (b) Power to purchase a commodity.

Thus Demand = Strong will + Purchasing power. But a third factor "price" cannot be ignored while discussing the concept of demand. As any change in the price of a good results a definite change in the demand for that good. Thus $q_d = f(p)$ (quantity demanded is a function of price).

According to Prof. Benham, "Demand is the quantity of a commodity which the consumers are prepared to purchase at a certain price during a specific period of time". Simply we can say that by demand we mean, the quantity of a commodity being purchased by any person at a specific price. Such as demand of 40 KG of wheat at Rs.25 per Kg. OR demand of 5 KG of apples at Rs.60 per KG etc.

LAW OF DEMAND

Law of demand can be described in view of the functional relationship between price and quantity demanded. Our common observation is that; with the change in the price of a commodity in the market its. Quantity demanded is inversely changed i.e. higher price less demand while more demand due to decrease in price. This behaviour of the consumer is called as "law of demand".

DEFINITION:

"other things remaining the same the demand of a commodity "expands" with the fall in its price while the demand "contracts" with the rise in price".

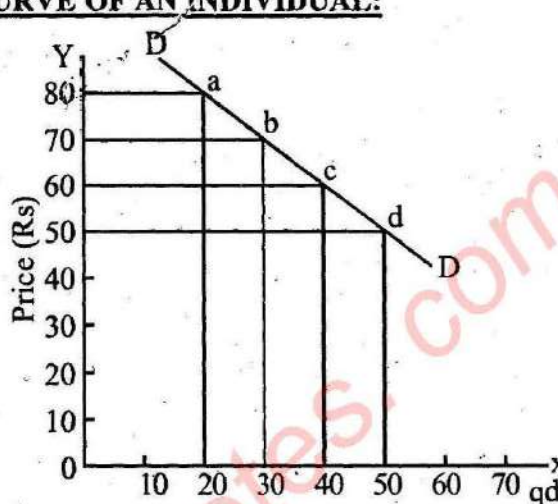
According to Professor Ferguson "There is an inverse relationship between the quantity demanded and price of a commodity, with the condition that; the income of the consumers and price of the substitutes remaining unchanged".

This law may be explained with the help of a schedule and diagram as under:

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DEMAND SCHEDULE AND DEMAND CURVE OF AN INDIVIDUAL:

Price of Ghee per K.G.	Quantity demanded (KG)
80 Rs.	20
70 Rs.	30
60 Rs.	40
50 Rs.	50



The above demand schedule shows the quantity of Ghee demanded by the consumer at different price levels. When the price is Rs.80 per KG quantity demanded is 20 KG. The demand goes on increasing with the decrease in the price level until it reaches upto 50 KG as the price of Ghee reaches to Rs.50 per KG

Demand Curve

In the diagram. We measure the quantity along x-axis and price along y-axis. Keeping in view various pairs of (qd) and (p) in the schedule different points such as a, b, c and d are taken. while joining these points together we arrive at DD demand curve. Which is negatively sloped. It shows inverse relationship between the price and the quantity demanded of Ghee

MARKET DEMAND CURVE

In the market there are many consumers demanding the commodity at different price levels. When we sum of the quantity demanded of the commodity at different prices of all the consumers in the market, we arrive at the "market demand curve" of the commodity.

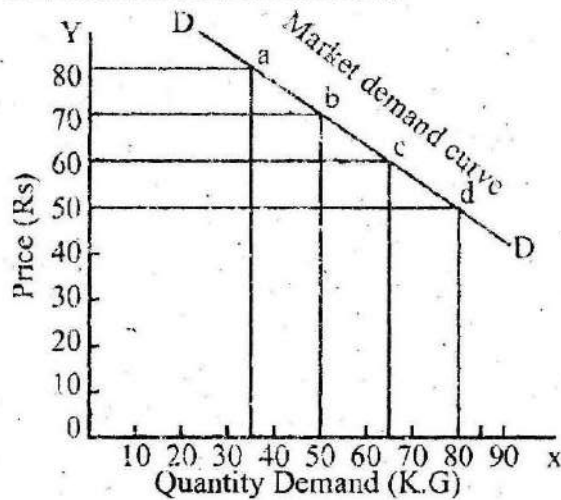
MARKET DEMAND SCHEDULE:

Price of Ghee per KG (Rs)	Consumer's Individual demand			Aggregate demand (KG)
	A	B	C	
80	5	20	10	35
70	8	30	12	50
60	10	40	15	65
50	15	50	20	85

It is assumed that there are three consumers in the commodity market, A, B and C. In view of the changes in the price of Ghee it's demand also varies but inversely. Thus a fall in price from Rs.80 to Rs.50 per KG results expansion in market demand from 35 (KG) to 85 (KG). It shows an inverse relationship between p & qd.

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MARKET DEMAND CURVE



In the diagram quantity demanded (Q_d) is measured along x-axis and price along y-axis. In view of various combinations of p and q_d in the schedule, points, a,b,c and d have been achieved. While connecting all these points DD Market demand curve has been drawn. DD moving from upward to downward is negatively sloped, just like the individual demand curve.

MAIN CHARACTERISTICS OF THE DEMAND CURVE

1. Demand curve describes the inverse functional relationship between the price and the quantity demanded of a commodity $q_d = f(p)$. Here, price is an independent variable while, quantity depends upon it.
2. Normally demand curve is negatively sloped and falls downward from left to right.

WHY DEMAND CURVE IS NEGATIVELY SLOPED?

There are following reasons behind it

(i) Entry of new buyers (price effect):

Fall in the price of a commodity attracts new buyers in the market. The people hesitating to purchase the commodity at higher prices, now start buying it after the fall in price resulting an increase in it's demand.

(ii) Income effect:

When the price of a commodity decreases, purchasing power of the consumer increases (resulting an indirect increase in real income). Thus more units are demanded at low price with the same amount of money.

(iii) Substitution effect:

When the price of required commodity falls, while the price of it's substitute remains unchanged, the consumers feel that this commodity is now relatively cheaper, therefore its demand increases while, on the other hand the demand for the substitute decreases.

(iv) Diminishing marginal utility:

According to the "law of diminishing marginal utility", every new unit gives less utility than the previous. So, additional units can be demanded only at lower price.

ASSUMPTIONS OF THE LAW OF DEMAND

Law of demand proves to be true if following conditions are fulfilled which are regarded as "Assumptions" of the law of demand.

1. No change in Consumer's Income

In order to prove the truth of the law of demand, it is assumed that there is no change

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in the income of the consumers. If, income changes, it may nullify the effect of change in price and consumers may buy more units at high price (due to increase in income) and less units at low price (due to decrease in income).

2. No change in consumers, Fashion, Tastes and Habits

The second assumption of the law of demand is that; the Fashion, Traditions, Tastes and Habits of consumers remain unchanged. If a good is "IN" with respect to "fashion" and "taste" it may command more demand even at a higher price on the other hand, the demand of a commodity falls even if, its price decreases.

3. No change in the number of consumers

It is also assumed that; the number of buyers of good in the market remains the same. Otherwise an increase in the number of consumers maybe resulting more demand either without change in the price level or even in increase in the prices. Similarly, a decrease in the number of consumers may result less demand even without change in the price level.

4. No change in the price of substitutes:

Law of demand will be true only if, the prices of substitute goods do not change. Because the prices of these goods influence the demand of the concerned commodity.

Thus buyers of wheat may increase it's demand even if it's price remains constant, but the price of rice has risen. (Rice is the substitute of wheat).

5. No change in the external value of money:

If people expect "devaluation of currency" in future, they may demand more even at high prices, because there is the danger of more increase in the price level after actual "devaluation of currency". Similarly, an expectation of "Revaluation" may discourage demand even at low prices. Thus the condition of "constant external value of money" is imposed to prove the truth of law of demand.

6. No discovery of new substitutes:

It is also assumed that; no discovery of new substitutes takes place. As the consumers are more attracted by the new products and the demand for old goods may contract without any rise in their price.

7. Homogeneity of all units of the good:

A very important assumption of the law of demand is that; all the units of the commodity should be equal in quality as well as in quantity. (Homogenous), it is a "universal truth" that the demand for a low quality unit of some good cannot be equal to the demand for a high quality unit at the same price OR a change in the price of good may not direct the demand of high and low quality unit in the same direction at the same ratio.

8. The good is not being used as a token of distinction:

The commodity under use must be normal and equally important for the consumers. The demand for a good being used as "token of distinction" OR "show off" just for superiority will not follow the "characteristics of the law of demand". The demand for such a good will expand more even at a higher price level.

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EXCEPTIONS AND LIMITATIONS

Under certain circumstances, the price and demand of the commodity move in the same direction. These are known as the exceptions and limitations of the law of demand.

1. Giffon goods:

A British Economist "Sir Robert Giffen" pointed some most inferior goods indicating a "positive functional relationship" between demand and price. The demand for these goods increases with the rise in their prices and vice versa. For instance, if we consider that Barley is an inferior good with respect to wheat. Then a decrease in the price of Barley will discourage its demand (The price effect being positive) while on the other hand the demand for wheat will increase at its same price (being superior to Barley).

2. Scarce goods:

When people expect that in future there will be shortage and scarcity in the supply of commodity and it will not be available in the market. They may purchase its more units even though price is rising. This is due to fulfill future requirements regarding the use of commodity.

3. The most precious goods:

The most precious goods i.e. Diamonds, Stones, Luxurious cars etc which are used as a "token of distinction" or a symbol of status are also known as the exceptions of the law of demand. The demand for such goods goes on increasing with the increase in their prices.

4. Emergency:

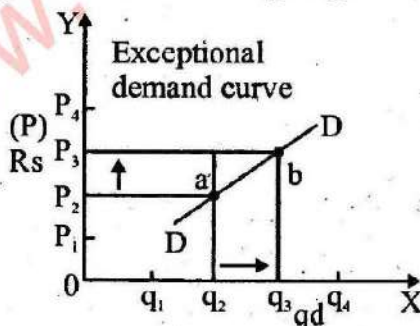
In view of the threat of war, natural calamities, worse law and order situation or emergency in the country, people start hoarding necessities of life and go on purchasing more quantity of goods even though price of such goods is increasing.

5. Ignorance and Carelessness:

Some people are ignorant with respect to the price and supply of goods in the market. So they may purchase more units of good at high price and less units at low price. Sometimes due to carelessness, the consumers do not bother about the high price and thus "dis-obey" the law of demand.

EXCEPTIONAL DEMAND CURVE

Keeping in view the above mentioned exceptions, the demand curve is positively sloped moving upward from left to right. It is regarded as "the exceptional demand curve" as shown by DD in the following diagram.



As price rises from P_2 to P_3 quantity demanded also increases from q_2 to q_3 . Thus DD "Exceptional demand curve" is positively sloped.

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Functional Equation of Demand

In view of "law of demand" the functional relationship between price and quantity demanded may be described as under.

The purchase plan of the consumer indicates that:

$$Y = p_1q_1 + p_2q_2 + p_3q_3 + \dots + p_nq_n$$

Here (i) y = the amount (part of income) consumer wants to spend

(ii) $p_1, p_2, p_3, \dots, p_n$ are the prices of different goods

(iii) $q_1, q_2, q_3, \dots, q_n$ represent the quantity demanded of these goods.

According to the law of demand, the determinants of demand are:-

(i) P = price of the commodity

(ii) N = Number of consumers

(iii) Y = Income of consumer

(iv) T = Taste, fashion, habits and traditions of consumer

(v) P_i = Price of substitutes

Thus .. $q_d = f(P, N, Y, T, P_i) \dots (1)$

(Quantity demanded is function of price, number of consumers, income, taste and price of substitutes)

It is assumed that in the short – run, N , T and P_i remain the same, thus in simple term

$$q_d = f(p, y) \dots (2)$$

Algebraically it is written as under

$$q_d = ap + by \dots (3)$$

Here q , p and y are variables, while a and b are co-efficients assumed as constants.

More simply it is also assumed that in the short period consumer's income also remains unchanged. Hence $by = c$ (constant)

Thus $q_d = ap + c \dots (4)$

As demand curve is negatively sloped, so, the co-efficient of p is always negative and standard form of demand equation:-

$$q = c - ap \dots (5)$$

If $c = 10$ and $a = -2$

Then $q = 10 - 2p \dots (6)$

Graphic Representation of Demand Function:

When $q = 10 - 2p$

Here p (price) is an independent variable. It may be given different values such as 1, 2, 3 4. To find the values of q , we put these values of p in the above mentioned function.

$$q = 10 - 2p$$

(i) $q = 10 - 2(1) = 10 - 2 = 8$

(ii) $q = 10 - 2(2) = 10 - 4 = 6$

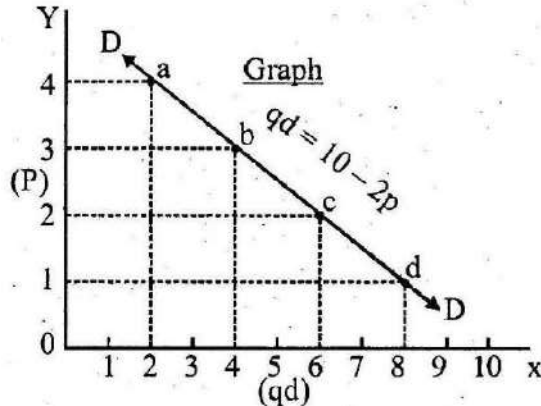
(iii) $q = 10 - 2(3) = 10 - 6 = 4$

(iv) $q = 10 - 2(4) = 10 - 8 = 2$

Schedule

p	1	2	3	4
q_d	8	6	4	2

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P is measured along y-axis and qd along x-axis.
 In view of schedule various combinations of p & qd are put in the graph getting points a, b, c and d. while connecting all these points we derive DD demand curve

DERIVATION OF DEMAND EQUATION

If demand schedule is given we can derive demand equation with the help of following formula:

$$q - q_1 = \frac{q_2 - q_1}{p_2 - p_1} (p - p_1)$$

Here q = symbol of quantity demanded
 q₁ = Initial quantity demanded
 q₂ = changed (2nd) quantity demanded
 p = symbol of price
 p₁ = Initial price
 p₂ = changed price.

According to the given schedule we derive demand equation.

	p_1	p_2	
P	5	10	15
qd	40	30	20
	q_1	q_2	

Putting the values into formula

$$q - 40 = \frac{30 - 40}{10 - 5} (p - 5)$$

$$q - 40 = -\frac{10}{5} (p - 5) = -2(p - 5)$$

$$q - 40 = -2p + 10 \text{ or } q = -2p + 10 + 40$$

$$q = -2p + 50 \text{ or } q = 50 - 2p$$

Thus the required demand equation is $q = 50 - 2p$

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SHIFT IN DEMAND CURVE

Demand for a commodity depends upon many factors, such as:

(i) Price of the commodity (ii) Income, Taste, Fashion, Habits, Traditions and Number of consumers (iii) Price of substitutes and complementary goods etc. A change in any of the above mentioned factors, results the change in the demand for the concerned commodity. According to the Economists, such a change in demand is classified as under:-

- (a) Change in demand due to change in the price. It is known as expansion or contraction in demand i.e. (movement along the same demand curve).
- (b) Change in demand due to other factors is, known as Rise or Fall in demand i.e. (shift in demand curve)

(a) Expansion and contraction of demand curve

OR

(Movement along the same demand curve)

When more quantity of the good is demanded due to decrease in the price of good, it is known as "Expansion" or "Extension" in demand.

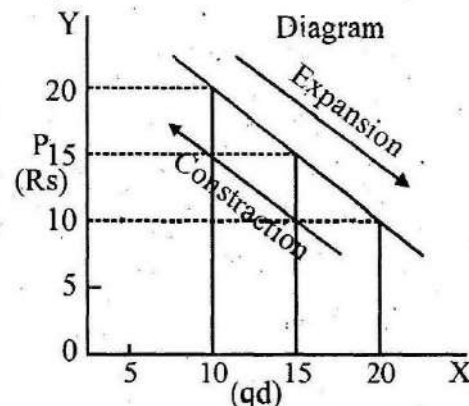
Under such a situation the movement in the schedule as well as along the same demand curve is from upward to down and left to right.

On the other hand if less quantity of the good is demanded due to increase in the price of the good, it is known as the "contraction in demand"; under these circumstances the movement in the schedule as well as in the diagram is indicated from downward to upward and right to left.

Both conditions are shown as under:-

price (P) Rs	Quantity Demand (qd) K.G
20	10
15	15
10	20

Contraction
Expansion



Right side arrow in the given schedule and diagram shows downward trend indicating an "Expansion or Extension" in demand while the left side arrow showing an upward trend indicates "Contraction in demand". This movement is along the same demand curve.

(b) Rise and fall in demand OR Shift in demand curve)

1. RISE IN DEMAND:

When demand for a commodity increases not due to change in it's own price, but due to change in "other factors" which are assumed to remain the same e.g. Y, N, T, Pi etc.

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It is known as “rise in demand” under these conditions new demand curve shifts “upward to the right of original demand curve”

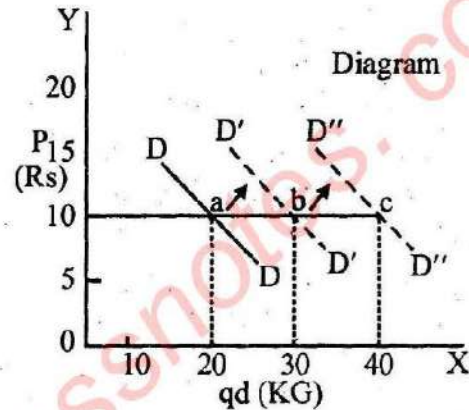
This upward shift in demand curve may be shown under two conditions.

- (i) Rise in quantity demanded at the same price.
- (ii) Quantity demanded remains the same at higher price.

RISE IN DEMAND

(i) More Qd at same price

P(Rs)	qd (KG)
10	20
10	30
10	40

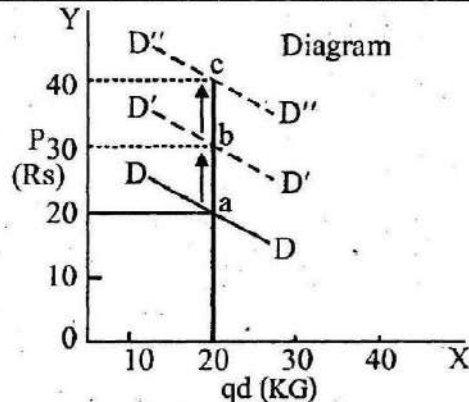


EXPLANATION:

DD is the original demand curve showing combination of P (Rs.10) and qd (20) in the schedule while price remaining the same with more qd (30) D'D' moves upward right to the original DD. While further increase in qd (40) at the same price helps to shift D''D'' above the D'D'.

(ii) Same Qd at higher price

P(Rs)	qd (KG)
10	20
20	20
30	20



EXPLANATION:

DD is the original demand curve. When price goes up (Rs.20) without any change in qd (due to the pressure of other factors) new demand curve D'D' moves upward. Similarly with the further rise in price (Rs.30). D''D'' shifts upward indicating “Rise in Demand”. The above mentioned “Two Ways” indicate “Rise in Demand”

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2. FALL IN DEMAND

When demand for a commodity decreases at the same price or fall in price shows no effect upon quantity demanded (qd) (due to the pressure of other factors assumed to remain the same), is known as "Fall in Demand".

Under these conditions new demand curve shifts "downward" as $D'D'$ and further as $D''D''$.

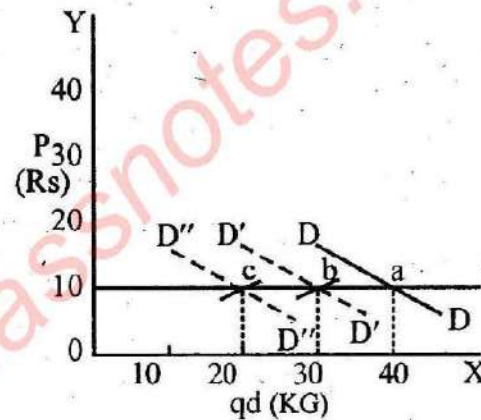
Two ways to explain it:

- Decrease in quantity demanded at the same price.
- Quantity demanded remains the same at lower price.

FALL IN DEMAND

(i) Less Qd at same price

P(Rs)	qd (KG)
10	40
10	30
10	20

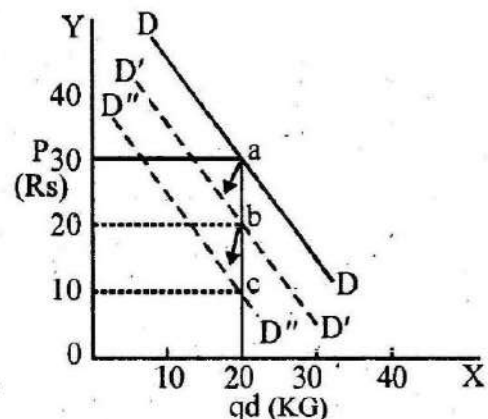


EXPLANATION:

DD is the original demand curve showing combination of P(Rs.10) and qd (40) at point a, price remaining the same with less qd (30) $D'D'$ moves downwardly left to the original DD. While further decrease in qd (20) at the same price results a shift of $D''D''$ downwardly.

(i) Same qd at lower price

P(Rs)	qd (KG)
30	20
20	20
10	20



EXPLANATION:

DD is the original demand curve showing combination of P(Rs.30) and qd (20) at point a-qd remaining the same even if price decreases at (Rs.20) new demand curve $D'D'$

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shifts downwardly. More decrease in P (Rs.10) at same qd (20) results a further downward shift in demand curve as D''D''.

These "Two ways" indicate the "Fall in demand".

CAUSES OF CHANGE IN DEMAND.

Note: "The concerned good is considered as a normal good".

Various factors are responsible for change in demand for a good. Some major among them may be discussed as under:

1. CHANGE IN THE PRICE OF THE GOOD.

According to the "Law of Demand" $q_d = f(p)$, thus the major factor influencing the change in quantity demanded is the change in price of the good concerned. It must be noted that change in quantity demanded and price of normal good is inverse (in opposite direction).

2. CHANGE IN THE INCOME OF CONSUMER:

Demand for a commodity is directly affected by the change in consumer's income. An increase in income enhances the purchasing power of consumers, thus they buy more of the commodity. While decrease in income discourages demand and the result is opposite.

3. CHANGE IN THE NUMBER OF CONSUMERS:

If number of consumers increases due to increase in population or any other factor, demand for goods will rise, while on the other hand shortage of consumers will be resulting a fall in the quantity demanded.

4. CHANGE IN THE HABITS AND FASHION:

If there is a "favourable" change in Fashion, preference or tastes of the consumers for a particular commodity, the demand will rise. On the other hand, if a commodity becomes "out of fashion" or the change in tastes is "unfavourable" towards that commodity, the demand will fall.

5. CHANGE IN THE PRICE OF SUBSTITUTES AND RELATED GOODS:

Demand for required good will be changed with the rise or fall in the price of the commodity substitutes and complements. For instance; rice is the substitute of wheat. An increase in the price of rice will cause rise in the demand for wheat and vice versa. Cars and petrol are jointly used (complements). Hence, an increase in the prices of cars will cause fall in the demand for both (cars and petrol).

6. CHANGE IN BUSINESS ACTIVITIES:

Favourable business conditions, expansionary phase of business cycle and better future prospects will encourage demand for goods in the markets, while slow business activities, recession and depression will definitely cause fall in the demand for goods.

7. CHANGE IN WEATHER CONDITIONS:

Weather conditions also influence the demand for goods. For instance; demand for ice, cold drinks and cold ice cream falls during cold winter season, while the demand for such goods rises during summer season and hot weather conditions.

8. CHANGE IN THE PSYCHOLOGY OF THE PEOPLE:

Simplicity, Austerity and self contentment discourages wasteful consumption and ostentation, resulting fall in demand while, under opposite conditions the demand for goods may move upward.

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9. CHANGE IN WEALTH AND INCOME DISTRIBUTION:

An equal distribution of income encourages the purchasing power of a common man. It strengthens the middle man as the income is transferred from the rich to the poor. Thus demand for consumer goods rises. While under un-equal distribution of wealth and income in the economy aggregate demand for consumer goods falls.

10. CHANGE IN MONEY SUPPLY:

Larger quantity of money and its circulation in the country encourages business activities resulting a rise in demand. But when money supply decreases, demand for goods also falls.

11. FISCAL POLICY:

More indirect taxes on consumer goods or direct taxes on consumers income will discourage the purchase and demand for goods. While tax concessions in this respect may establish a positive effect upon demand, sale and purchase of goods.

12. PUBLICITY AND ADVERTISEMENT:

Advance electronic and print media can play a major role through advertisement and publicity in changing the taste, choice, preferences and hence demand for goods. A positive and effective advertisement campaign may attract more people to purchase goods resulting rise in demand.

13. CHANGE IN SAVING:

If due to different reasons people become more conducive towards saving, consumption expenditure will go down resulting fall in demand for goods and vice versa.

DIFFERENT CONCEPTS OF DEMAND

Some major concepts of demand may be discussed as under:

1. DERIVED DEMAND:

Demand for some goods is not direct, rather it is derived from the demand for some other goods. For instance, demand for a house is direct but for the construction of that house demand for bricks, cement, iron, wood, masson, labour etc. is indirect and derived from the demand for that house. Similarly, demand for the factors of production is derived from the demand for consumers goods.

2. JOINT DEMAND:

The satisfaction of some human wants depend upon the simultaneous demand for more than one goods i.e. car and petrol, pen and ink, bat, ball, ground and wills, services of empire etc. such goods have joint demand and are known as "complementary goods". Change in the price of one good definitely influences the demand for other good.

3. COMPOSITE DEMAND:

Some goods are used for different purposes. For instance, Gas may be demanded for (i) House hold consumption (ii) Industrial purposes (iii) Commercial concerns etc. The sum total of demand for gas in every concern becomes it's "composite demand" Increase in demand for one purpose results less quantity available for the other. Thus these concerns "fight against each other" to fullfil their own demand.

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ELASTICITY OF DEMAND

CONCEPT OF ELASTICITY:

The quantities changing their values size or structure are known as variables. This characteristic of change is regarded as "elasticity".

In economic theory the concept of elasticity is of paramount importance. As change in the value of one or more variables, effects the values of many other variables. This type of change leaves negative or positive effects upon different "groups of society" and different "sectors of the economy", i.e. $q_d = f(p, y, p_i)$ and $Q = f(L, N, K, O)$ etc.

ELASTICITY OF DEMAND:

According to "law of demand" it is clear that a change in price results an inverse change in quantity demanded. An increase in price shows "contraction in demand" and vice versa.

The concept of elasticity was initially introduced by Alfred Marshall in his well-known book "Principles of Economics".

"Elasticity of demand is the rate of change in quantity demanded in response to changes in price. He further says," the elasticity for responsiveness of demand in a market is greater or smaller according as the amount demanded increases much or little for a given fall in prices and diminishes much or little for a given rise in price".

According to Stonier and Hague "Elasticity of demand is a technical term used by economists to describe the degree of responsiveness of the demand for a good to the change in its price".

Simply we can say that "price elasticity of demand" shows "the sensitivity of quantity demanded of a good to change in its price".

The study of Economics theory tells us that the demand for all goods is not equally elastic. It may be as under:-

1. More elastic demand
2. Less elastic demand
3. Zero elasticity of demand (perfectly in-elastic)
4. Infinite elasticity (perfectly elastic)

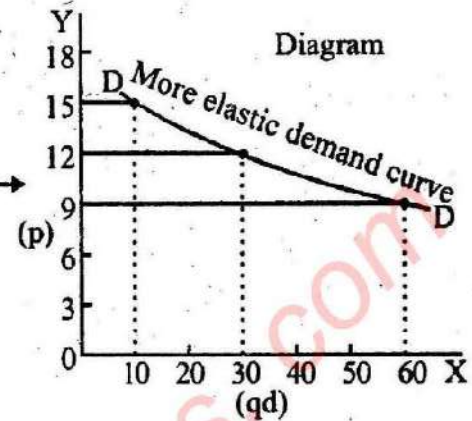
1. MORE ELASTIC DEMAND:

A relatively smaller change in price resulting a big change in quantity demand for a good is known as "more elastic demand good". Luxuries i.e. colour T.V., Air Conditioners, costly cars etc fall in this category. More elastic demand curve falls slowly and becomes more flatter shown as under:

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Schedule

Price of colour T.V. (Thousand Rs.)	qd
15	10
12	30
9	60

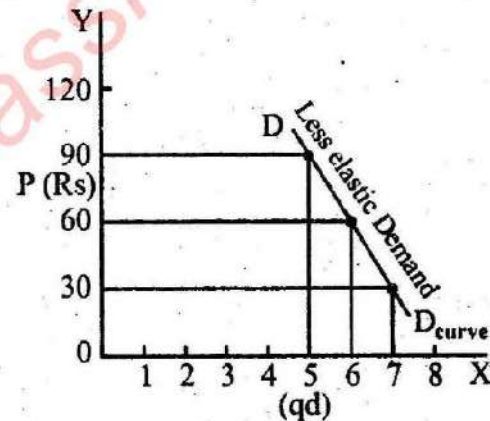


2. LESS ELASTIC DEMAND:

A relatively large change in price resulting a smaller change in quantity demanded for a good is regarded as "less elastic demand good". Basic necessities i.e. Flour, Ghee, Salt, Vegetables etc. become under this category. More elastic demand curve falls sharply downwardly and becomes steeper shown as under:

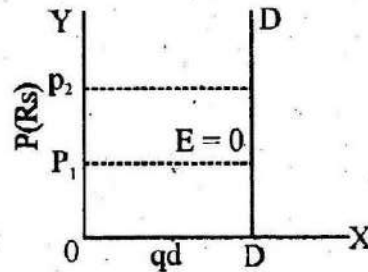
Schedule

Price of Ghee (Rs. Per KG)	qd (K.G)
90	5
60	6
30	7



3. ZERO ELASTICITY OF DEMAND

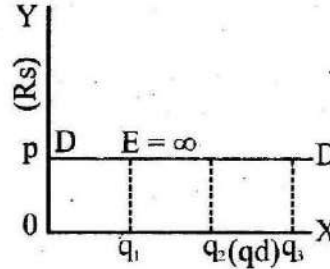
When demand does not respond even a big change in Price. Demand curve becomes vertical towards x-axis.



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4. INFINITE ELASTICITY

When demand goes on increasing at the same price. Demand curve becomes perfectly elastic and parallel to x-axis.



MEASUREMENT OF ELASTICITY OF DEMAND

There are three methods to measure the elasticity of demand.

1. Unitary Method OR Total Outlay Method.
2. Flux Method OR Percentage Method
3. Formula Method.

1. UNITARY METHOD OR TOTAL OUTLAY METHOD.

According to this method $E_d = \frac{\text{Proportional Change in quantity demanded}}{\text{Proportional change in Price}}$

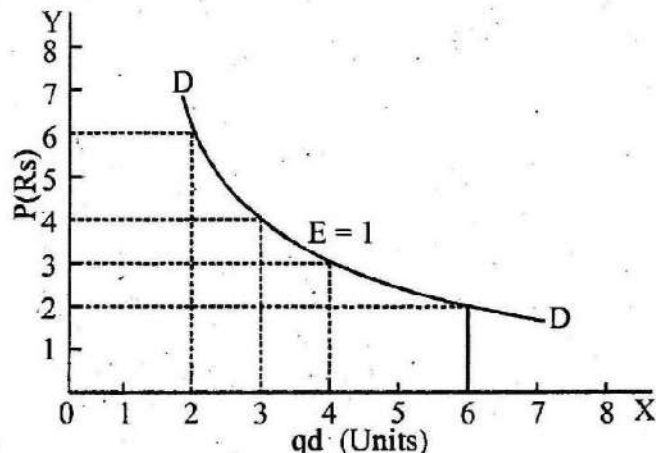
It may take three forms:-

- (a) $E = 1$ (b) $E > 1$ (C) $E < 1$

(A) ELASTICITY OF DEMAND EQUAL TO UNITY $E = 1$

When the change in Price and demand for a good is proportionate. The rate of change in both being equal indicates that with the increase or decrease in price, the total outlay remains unchanged. These goods are neither included in "necessaries" nor in "Luxuries" i.e. Tissue Paper, Paper weight etc.

Schedule		
Price (Rs)	qd (units)	Total outlay p.qd
6	2	12
4	3	12
3	4	12
2	6	12



DD looks like Rectangular hyperbola and indicates that $E = 1$.

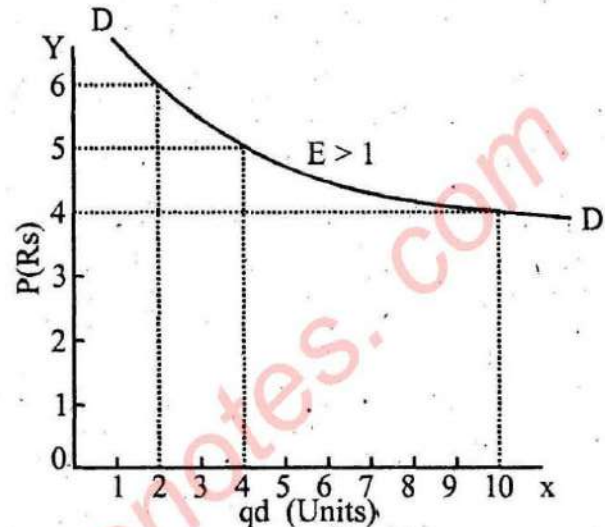
(B) ELASTICITY OF DEMAND GREATER THAN UNITY, $E > 1$

When proportionate change in demand for a good is great than the change in it's

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price. The total outlay increases with the decrease in price and vice versa. Luxuries are included in this category.

Schedule		
Price (Rs)	qd (units)	Total outlay (p.qd)
6	2	12
5	4	20
4	10	40

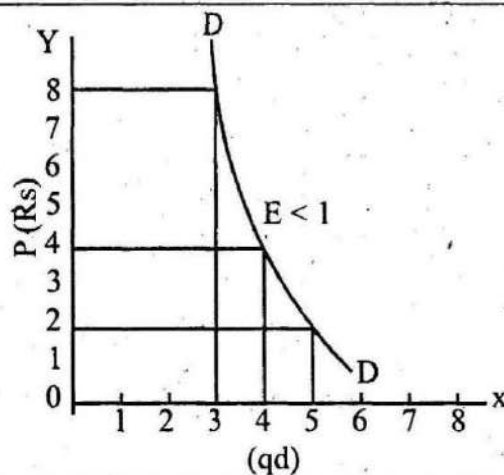


(C) ELASTICITY OF DEMAND LESS THAN UNITY $E < 1$

When the rate of change in price is greater than the proportionate change in the quantity demanded of a good its $E < 1$. Under such a situation total outlay increases with the increase in price and vice versa. Necessaries fall in this category.

Price	qd (Units)	Total outlay P.qd
2	5	10
4	4	16
8	3	24

A steeper DD indicates that $E < 1$



2. FLUX METHOD OR PERCENTAGE METHOD

This method has been introduced by Flux. According to it: -

$$E_d = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}}$$
 when there is an equal % change in price and quantity demanded for a good such as 10% increase in price resulting 10% decrease in quantity demanded then $E_d = \frac{10}{10} = 1$ ($E = 1$). If % change in quantity demanded is greater than the % change in price i.e. 10% increase in price decreasing quantity demanded by 20%

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then $E_d = \frac{20}{10} = 2 (E > 1)$.

When % change in price is greater than the % change in quantity demanded for a good i.e 10% increase in price indicating 5% decrease in quantity demanded. Then $E_d = \frac{5}{10} = 0.5 (E < 1)$.

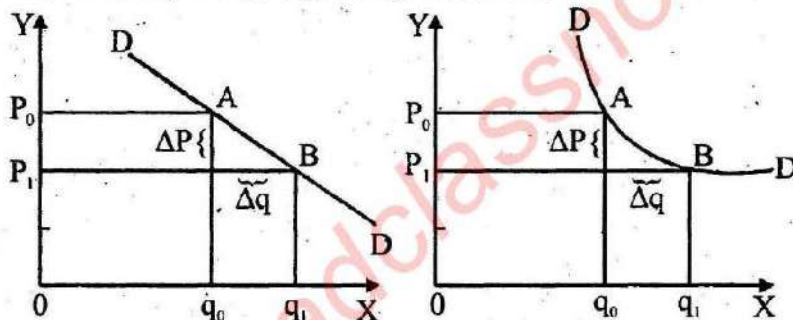
3. FORMULA METHOD:

Elasticity of demand may be measured according to the following formulas:

- Arc elasticity formula
- Point elasticity formula.

Arc elasticity formula:

When there is prominent and major change in price and quantity demanded of a commodity. Elasticity between two distinct points (A-B) of a demand curve can be measured with the help of Arc elasticity formula given by R.G.D. Allen.



Formula

$$E = \frac{\frac{q_0 - q_1}{q_0 + q_1}}{\frac{p_0 - p_1}{p_0 + p_1}}$$

E = Co-efficient of elasticity
 q_0 = Original quantity
 q_1 = Changed quantity
 p_0 = Original price
 p_1 = Changed price

Schedule

	p_0	p_1
P	25	20
q	8	10
	q_0	q_1

$$E = \frac{\frac{8-10}{8+10}}{\frac{25-20}{25+20}} = \frac{\frac{-2}{18}}{\frac{5}{45}} = \frac{-2}{18} \times \frac{45}{5} = -1 (E = 1)$$

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Here $q_0 = 8$, $q_1 = 10$, $p_0 = 25$, $p_1 = 20$

(ii) Point elasticity formula:

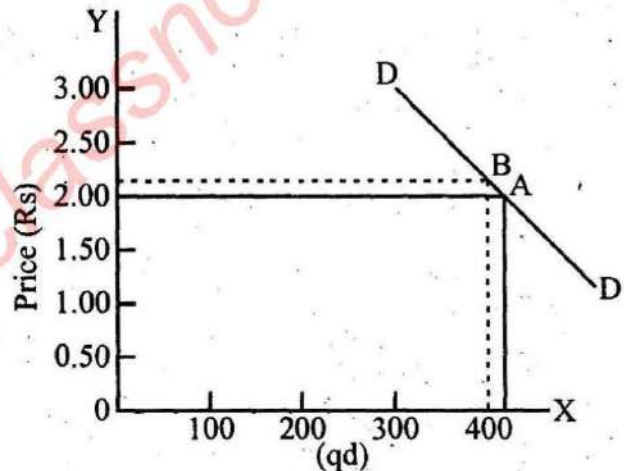
This formula has been given by Alfred Marshall. It is used to measure the elasticity of demand when there are very minor changes in price and quantity demanded for a good.

$$\text{Formula} \quad E = \frac{\frac{\Delta q}{q}}{\frac{\Delta p}{p}} \quad \text{OR} \quad \frac{\Delta q}{q} \times \frac{p}{\Delta p} \quad \text{OR} \quad \frac{p}{q} \cdot \frac{\Delta q}{\Delta p}$$

Here q = original quantity demanded
 Δq = change in quantity demanded
 p = original price
 Δp = Change in Price

Schedule

Price (P) Rs	Quantity demanded (q/d)
2.00	400
2.01	398
$\Delta p = -0.01$	$\Delta q = 2$



Here $q = 400$, $\Delta q = 2$
 $P = 2$, $\Delta p = -0.01$

Putting the values in Formula

$$E = \frac{2}{400} \times \frac{2}{-0.01} = \frac{2}{400} \times \frac{200}{-1} = -1$$

Thus $E = 1$

GEOMETRIC APPROACH

Under this approach elasticity at a point of demand curve is measured by comparing the distance from the point downward toward x-axis by the distance upward to the y-axis.

For instance in the following diagram we want to measure elasticity of demand at point E of DD

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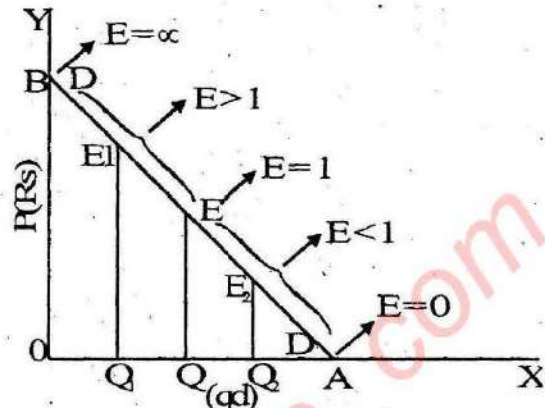
Here $E = \frac{EA}{EB}$. Because the distance between

$EA = EB$

Thus at point E, elasticity ($E = 1$)

At point E_1 , $E = \frac{E_1A}{E_1B} = E > 1$

At point E_2 , $E = \frac{E_2A}{E_2B} = E < 1$

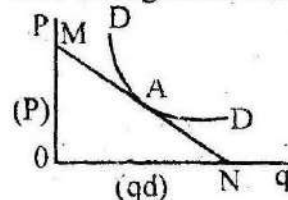


At point A, $E = 0$

While at point B, $E = \infty$

Thus $E = \frac{\text{Length of lower part}}{\text{Length of upper part}}$ of DD curve

If the demand curve is not a straight line, but a curvature shown as under, then we draw a tangent MN along DD. Here slope of the tangent indicates the elasticity.



At point A, $E = \frac{AN}{AM} = 1$, $AN = AM$ (Equal distance)

OTHER CONCEPTS OF THE ELASTICITY OF DEMAND

(A) INCOME ELASTICITY OF DEMAND

Consumer's income is one of the important determinants of demand. Thus any change in the income of consumers (y) will effect demand and it will also change.

So, we can say, "Income elasticity of demand is the degree of responsiveness of demand to the change in consumer's income."

Formula $E_y = \frac{\frac{\Delta q}{q}}{\frac{\Delta y}{y}}$ OR $\frac{\Delta q}{q} \div \frac{\Delta y}{y} = \frac{\Delta q}{q} \times \frac{y}{\Delta y} = \frac{y}{q} \times \frac{\Delta q}{\Delta y}$

Here, y = Consumer's Income,
 Q = Original quantity

Δy = Change in Consmer's Income

Δq = Change in quantity

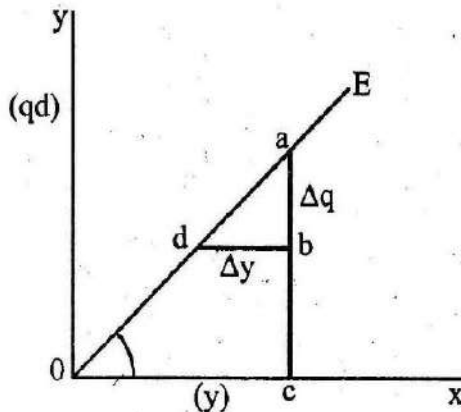
Δy putting the values in the formula

Y	500	600	100
Qd	20	25	5

↑
 Δq

$$E_y = \frac{\frac{5}{20}}{\frac{100}{50}} = \frac{5}{20} \times \frac{500}{100} = \frac{5}{4} = 1 \frac{1}{4} \text{ or } 1.25 > 1$$

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Main points

1. Ey of normal goods is positive
2. Ey of inferior goods is negative.
3. $E_y \leq 1$ (equal, greater or less than unity)
4. With respect to luxuries $E_y > 1$
5. With respect to necessities $E_y < 1$
6. When Δy does not effect q then, $E_y = 0$

In the above diagram $E_y = 1$

Because proportionate change in Income (Δy) and quantity (Δq) is equal.

(B) CROSS ELASTICITY OF DEMAND

Under cross elasticity of demand (E_c) change in the price of commodity (B) results a change in the demand for commodity (A) while the price of (A) remains the same. Thus we can say that "the rate of responsiveness of quantity demanded of commodity A to changes in price of commodity B" is regarded as cross elasticity of demand.

For instance, wheat and Rice are substitutes. An increase in the price of Rice will affect demand for wheat and it will go up.

Similarly, petrol and cars are complements (jointly demanded) A rise or fall in the price of cars will lead to an inverse change in the demand for petrol..

$$\text{Formula } EC = \frac{\frac{\Delta q_A}{q_A}}{\frac{\Delta p_B}{p_B}}$$

Here EC = Cross elasticity of demand
 q_A = Quantity demanded for good A
 Δq_A = Change in quantity demanded for good A.
 p_B = Price of good B
 Δp_B = Change in the price of good B.

Commodities	Price (Rs) for 40 KG	Quantity demanded (KG)	
Wheat A $P_A \rightarrow$	100	40 q_A	
		$\frac{50}{10} = \Delta q_A$	
Rice B $P_B \rightarrow$	2000		
	$\frac{2500}{500} = \Delta p_B$		

Putting the values in the formula:

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$$EC = \frac{\frac{10}{40}}{\frac{500}{2000}} = \frac{10}{40} \times \frac{2000}{500} = 1 \text{ (equal to unity)}$$

In the above statement P_B (price of Rice) has gone up ($\Delta P_B = 500$) But good A (Wheat) being substitute has shown an increase in its demand ($\Delta Q_A = 10$)

FACTORS INFLUENCING THE ELASTICITY OF DEMAND

The following factors influence the elasticity of demand whether it is "more" or "less elastic".

1. NATURE OF GOOD:

The nature of different goods differ from each other. Thus their elasticity of demand also differs

- The elasticity of demand for necessities i.e. Flour, Ghee, Salt etc. is less elastic. Because a greater change in their price results a very minor change in their demand on the other hands luxuries. costly cars, colour TV etc. are more elastic goods. Because a little change in price causes a greater change in their demand.
- Perishable goods i.e. Vegetables, Fruits, Meat etc. are less elastic goods while durable goods i.e. Motor Bike, Sofa Set, wall clock etc. are more elastic goods. Demand for such durable goods may be postponed for sometime in view of a rise in their prices.

2. PRICE LEVEL:

Most precious goods i.e. Diamonds, Jewellery, B.M.W. Cars etc. and Giffen goods (most inferior goods) having a very low price i.e. plastic shoes, lunda clothes are less elastic or inelastic goods. Because a rise or fall in their prices do not affect demand for Richer or the poorer.

3. DIFFERENT USES OF GOODS:

A good having only one specific use is less elastic. But goods having many uses have more elastic demand.

These goods / services may not be used for less important purposes if their price increases. For instance, Natural gas is used for (i) household services i.e. heating, cooking, lighting etc. (ii) Commercial purposes and (iii) Production purposes through industrial units and power generating plants, If there is a quick and sudden rise in its price, it may not be used for heating and lighting. Demand for Gas is thus elastic for some uses and inelastic for others.

4. SUBSTITUTE FOR THE GOOD:

Availability of close substitute of a good in the market, makes its demand more elastic. If the price of such a good rises, people start purchasing its substitute e.g. if price of Coca Cola rises, people start using pepsi.

5. ABILITY OF THE GOOD TO POSTPONE IT'S USE:

The goods whose demand can be deferred for sometime are known as "more elastic goods." Because these are highly responsive to the change in their prices. If price rises, consumer's can wait for sometime e.g. T.V., Radio, Furniture.

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6. FASHION, HABIT AND ATTITUDE:

When some commodity "comes into Fashion" or "form a part of habits and tastes" becomes less elastic or inelastic. People do not care about it's price and prefer it's purchase even at a higher price.

7. JOINTLY DEMANDED GOODS

The goods which are jointly demanded are less elastic. As a rise or fall in their prices may not influence their demand largely unless the prices of complementary goods also change e.g. demand for cars and petrol, hockey stick and ball. A fall in the price of petrol will have a very minor effect unless the price of cars also fall.

8. RELATIVE IMPORTANCE IN HOUSEHOLD CONSUMPTION:

Goods having a very little share in the household budget e.g. Salt, Chillies, etc. are "less elastic" because a rise or fall in their prices do not upset the budget estimates at large. While the goods having a large share in the budget i.e. Flour, Gee, Milk, etc. are "more elastic goods."

9. LEVEL OF INCOME:

Income elasticity of demand for goods of rich people is very low because they demand according to their requirements. Marginal utility of money for them is also very low. On the other side; income elasticity of demand of poor is normally high. They immediately decrease their demand with falling income.

PRACTICAL IMPORTANCE OF ELASTICITY OF DEMAND

Importance of elasticity of demand in practical life may be discussed as under:

1. GUIDANCE TO FINANCE MINISTER:

The responsibility of Budget preparation lies on the shoulders of Finance Minister. For this purpose he has to collect Revenue to meet the expenditure items. Tax being the major source of revenue must be imposed keeping in view demand elasticity of various goods. Inelastic and less elastic goods can be taxed heavily. But heavy taxes on more elastic demand goods may result a fall in revenue. Thus a wise step in this respect may be more fruitful and result oriented.

2. GUIDANCE FOR ENTREPRENEURS AND PRODUCERS:

These people participate in the production process. Their main aim is to maximise profit. For this purpose to increase the demand of the product, the price may be kept low with respect to more elastic demand goods. On the other hand, inelastic and less elastic demand goods may give more profit even if at a high price. Thus demand elasticity of products plays an important role at the time of fixation of prices by the entrepreneurs and producers.

3. GUIDANCE FOR MONOPOLIST:

Although a Monopolist has full control over the production or supply of the commodity. Moreover there is no close substitute of the product. Even then the price elasticity of demand plays an important role for profit maximization. If he finds that the demand is elastic, a lower price results a larger demand. While a high price for inelastic and less elastic goods has a very little effect upon quantity sold. "Price discrimination policy" may also be adopted in view of difference in the elasticity of demand for the product.

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4. GUIDANCE FOR FARE DETERMINATION IN TRANSPORT SECTOR:

Demand elasticity also plays an important role to determine the fares of different means of transportation. The fare of Air Conditioned and luxury coaches is kept high in view of the spending capacity of rich people and being less elastic for them. On the other hand, the fare of ordinary coaches and 2nd class railway compartments is fixed at a lower level, because of having elastic demand for masses. Moreover, the fare for huge size and weight commodities having relatively less values e.g. Bricks, Fodder, Straw is kept low in view of their demand elasticity.

5. GUIDANCE FOR PRICE DETERMINATION OF JOINT PRODUCTS:

Prices of jointly produced goods are determined by their relative elasticities of demand. The price will be higher for the product having inelastic or less elastic demand than for the other. Wheat and straw are jointly produced. The price of wheat is higher than the relative price of straw, because the demand for wheat is less elastic, while straw has more elastic demand.

6. HELPFUL IN FOREIGN EXCHANGE EARNINGS:

Demand elasticity for exports also plays an important role in foreign exchange earnings. As the price and export duty may be kept higher for goods having less elastic demand. While a relative low price and duty exemption is more useful for more elastic demand goods to enhance the demand for exported goods in their foreign markets.

7. EXCESS PRODUCTION OF CERTAIN GOODS

When the demand for product is inelastic its more production may result in small revenues. Thus in view of growers, plenty or excess production of such goods shows a decrease in total income rather than increase, which means poverty.

8. DEVALUATION OF CURRENCY:

Devaluation of local currency can only be fruitful if our demand for imports and foreign demand for our exports is more elastic. Because the demand for export may be increased while demand for imports may be decreased to earn net benefit only under above mentioned conditions.

OBJECTIVE QUESTIONS

Select and tick (✓) Correct answer.

- Functional relationship between demand and price $q_d=f(p)$ is known as:
(a) Demand schedule (b) Law of demand
(c) Desire for demand (d) Income elasticity
- Decrease in demand due to rise in price is regarded.
(a) Fall in demand (b) Rise in demand
(c) Contraction in demand (d) Expansion in demand
- What type of functional relationship is between p and q_d .
(a) No relationship (b) Positive (c) Inverse (d) Increasing

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4. Which formula is used to measure elasticity of demand with minor changes in p and qd .
(a) Arc elasticity (b) Cross elasticity
(c) Income elasticity (d) Point elasticity
5. Find out the demand equation as under:
(a) $q = 10 + 5p$ (b) $q = -p + 5$
(c) $c = a + by$ (d) $q - 128 - 9p = 0$
6. When price (P) rises from Rs.5/- to Rs.6/- per unit and corresponding change in qd is 10 units from 20 units, $EP = ?$
(a) 0.8 (b) 1 (c) 1.5 (d) -2.5
7. When 5% change in p results 10% change in qd . Ed will be:
(a) $E = 1$ (b) $E < 1$ (c) $E > 1$ (d) $E = 0$
8. Why shifting in demand curve takes place.
(a) Change in Habits and customs (b) Change in Income
(c) Change in price (d) Change in (a) and (b)
9. How demand for a Giffen good will be affected due to fall in its price.
(a) qd will increase (b) qd will decrease
(c) No change (d) Infinite elastic
10. Following formula $E = \frac{\frac{\Delta q}{q}}{\frac{\Delta p}{p}}$ to measure Ed was introduced by?
(a) James Tobin (b) Prof. Robbins
(c) Alfred Marshall (d) Prof. Allen
11. The demand for goods having close substitutes is...?
(a) Less elastic (b) More elastic
(c) Perfect elastic (d) Zero elastic
12. Which good is having elasticity more than unity...?
(a) Salt (b) Ghee (c) Car (d) Flour
13. When total outlay remains the same due to change in the price of a good. Its Ed will be...?
(a) $E > 1$ (b) $E = 1$ (c) $E < 1$ (d) $E = \infty$
14. When total outlay rises with the rise in price, the nature of Ed will be ...?
(a) $E = 1$ (b) $E < 1$ (c) $E > 1$ (d) $E = 0$
15. Which of the following are not jointly demanded....?
(a) Horse and Cart (b) Car and Petrol
(c) Shoes and Spectacles (d) Pen and Ink.

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16. The slope of an exceptional demand curve is ...?
(a) Positive (b) Negative (c) Horizontal (d) Vertical
17. Aggregate demand for electricity with respect to Household, Commercial and Industrial uses is known...?
(a) Joint demand (b) Composite demand
(c) Derived demand (d) Implicit demand
18. In spite of decrease in price when demand remains the same it is called...?
(a) Rise in demand (b) Expansion in demand
(c) Fall in demand (d) Contraction in demand
19. The demand for goods whose uses are postponed is regarded....?
(a) Zero elastic (b) Less elastic
(c) More elastic (d) Infinite elastic
20. Arc elasticity formula was presented by....?
(a) Alfred Marshall (b) Adam Smith
(c) M. Friedman (d) Prof. Allen

DESCRIPTIVE PART - I

SHORT NOTES

1. Explain the concept of demand.

According to Prof. Benham, Demand is the quantity of a commodity which the consumers are prepared to purchase at a certain price during a specific time period." Such as 5 K.g Mango at P=60 per K.G etc.

2. Define "Law of Demand".

"Other thing remaining the same the demand of a commodity expands with the fall in its price while the demand contracts with the rise in price".

P.	Qd
50	40
60	30
70	20

3. Write down four assumption of "law of Demand"

- No change in consumer's income.
- No Change in fashion, taste and habits.
- No change in number of consumers.
- No Change in price of Substitutes.

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

4. Why demand curve is negatively slopes?

- (i) Due to entry of new buyers price effect,
- (ii) due to income effect,
- (iii) substitution effect,
- (iv) due to diminishing MU.

5. Find out the "exception" of law of demand?

Giffon goods, scarce goods, very precious goods In case of emergency, Ignorance and carelessness.

6. What is the difference between following concepts.

Joint demand: Some human wants are satisfied by more than one goods. Such goods have joint demand e.g. car and petrol, pen and ink, bat and ball.

Composite demand: Some goods are used for different purposes for example, gas, coal etc. The total demand for gas or coal shows its composite demand.

7. Explain the concept of "Derived demand"?

Some goods are indirectly demanded e.g. demand for house in direct but for it when we demand bricks, cement, labor, iron, wood etc. are indirectly demanded is called derived demand.

8. What is meant by "Elasticity of demand"?

According to Alfred Marshall:

"Elasticity of demand is the rate of change in quantity demanded in response to changes in price" suppose price of x rises from Rs. 10 to Rs.12 unit and demand contracts from 20 units to 12 units.

$$Ed = \frac{P}{q} = \frac{dq}{dp}$$

9. Write down four factors indicating "Practical Importance" of elasticity of demand?

- (i) Guidance to finance minister
- (ii) Guidance for entrepreneurs and producers
- (iii) Guidance for monopolist
- (iv) Determination of price in transport sector

10. Write down following formulas.

a) Point Elasticity $Ed = \frac{P}{q} = \frac{\Delta q}{\Delta p}$

b) Arc Elasticity

$$Ed = \frac{\frac{q_0 - q_1}{q_0 + q_1}}{\frac{P_0 - P_1}{P_0 + P_1}}$$

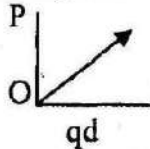
ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

11. Point out four factors influencing elasticity of demand?

- (i) Nature of the goods
- (ii) Price level
- (iii) Different uses of goods.
- (iv) Substitutes for the good.

12. What is an exceptional demand curve?

An exceptional demand curve is positively sloped Demand curve moving upward left to right which shows an increase in demand as price increases and vice versa. Its slope is



13. Write down the "functional equation of demand"

$Q_d = f(P)$ whereas assuming other determinants constant i.e. N, Y, T, P_i , $q = c - ap$ whereas; standard form is $q = 10 - 2p$

14. What is cross elasticity of demand?

Change in price of 'B' commodity, resulting of change of demand for 'A' commodity while, the price of 'A' remains the same is called cross elasticity of demand.

15. Bring out difference between "Price Elasticity (EP) and income elasticity (EY) of demand?

Income elasticity of demand is the degree of responsiveness of demand to the change in consumer's income" on the other hand, "Price elasticity of demand is the ratio of change in demand due to change in Price.

$$EP = \frac{\text{Proportionate Change in Demand}}{\text{Proportionate change in Price}}$$

PART - II

1. Explain the "law of demand" with the help of schedule and diagram.
2. Derive curves to differentiate between "movement along the demand curve" and "shifting of demand curve".
3. What is elasticity of demand. How it is measured?
4. Bring out difference between a "Normal demand curve" and "exceptional demand curve".
5. Derive the equation of "demand function" in view of following data.

p	2	3
q	6	4

